

THE MULTIVERSE VIA MULTIPLE TRACES

$$\tau_\alpha(e_n) = n^{-\alpha} \cdot \text{Each } \alpha \text{ is a Universe} \cdot \text{RH is the Universal Law}$$

Anthropic Warrior

Chien Dich Riemann Hypothesis - Campaign Document dv236

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Abstract

The Memory Framework A_L carries a canonical trace $\tau(en) = 1/n$. But the algebra A_L admits a continuous family of traces $\tau_\alpha(en) = n^{-\alpha}$ for $\alpha > 1$. Each such trace defines a distinct physical universe with partition function $Z_\alpha(\beta) = \zeta(\alpha+\beta)$, distinct fine structure constant, mass gap, and Hagedorn temperature. Our universe corresponds to $\alpha = 1$. The Riemann Hypothesis — that all non-trivial zeros of $\zeta(s)$ have $\text{Re}(s) = 1/2$ — is invariant across all values of α : it is a universal law of the multiverse, not a property of any particular universe. We develop this multiverse picture rigorously, classify the universes by their spectral properties, and show that the anthropic principle selects $\alpha = 1$ as the unique value supporting complex observers.

MSC 2020: 46L36, 11M26, 83F05 (Cosmological models), 83E30.

1. The Family of Traces τ_α
2. Each α is a Universe: Physics from τ_α
3. The Invariant: RH Across All Universes
4. Anthropic Selection: Why $\alpha = 1$
5. Phase Transitions Between Universes
6. The Multiverse Partition Function
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1. The Family of Traces τ_α

The hyperfinite II_1 factor R has a unique trace up to normalization. The algebra $A_L = l^2(\mathbb{N}, n^{-1})$ carries a canonical trace $\tau(e_n) = 1/n$ corresponding to $\alpha = 1$. However, if we allow the underlying measure on $\mathbb{N}$ to vary, we obtain a continuous family:

Definition 1.1 (The α -Family of Traces)

For $\alpha > 1$, define the α -trace on A_L by: $\tau_\alpha(e_n) = n^{-\alpha} / \zeta(\alpha)$ [normalized so that $\tau_\alpha(1) = 1$] The corresponding Hilbert space: $X_{\{D-E\}^\alpha} = l^2(\mathbb{N}, n^{-\alpha})$ Partition function: $Z_\alpha(\beta) = \tau_\alpha(e^{-\beta H}) = \sum_n n^{-\alpha} * e^{-\beta \log n} = \sum_n n^{-(\alpha+\beta)} = \zeta(\alpha+\beta) / \zeta(\alpha)$ Our universe: $\alpha = 1$. Then $\tau_1 = \tau$ (the canonical trace).

Theorem 1.2 (Each τ_α is a Valid Trace [TIGHT])

For each $\alpha > 1$, τ_α satisfies all trace axioms: (i) Linearity: $\tau_\alpha(\lambda a + b) = \lambda \tau_\alpha(a) + \tau_\alpha(b)$ (ii) Tracial: $\tau_\alpha(ab) = \tau_\alpha(ba)$ for all a, b (iii) Positivity: $\tau_\alpha(a^*a) \geq 0$ (iv) Normalization: $\tau_\alpha(1) = 1$ Moreover, for $\alpha \neq \alpha'$, τ_α and $\tau_{\alpha'}$ are inequivalent: there is no unitary U in A_L such that $\tau_\alpha(\cdot) = \tau_{\alpha'}(U \cdot U^*)$.

Proof. (i)-(iv) are direct verification. Inequivalence: if $\tau_\alpha(a) = \tau_{\alpha'}(U a U^*)$ for all a , then in particular $\tau_\alpha(e_n) = \tau_{\alpha'}(U e_n U^*)$. But $\tau_\alpha(e_n) = n^{-\alpha} / \zeta(\alpha)$ and $\tau_{\alpha'}(e_n) = n^{-\alpha'} / \zeta(\alpha')$. These have different asymptotic growth: $n^{-\alpha}$ vs $n^{-\alpha'}$. No unitary can change the asymptotic growth of eigenvalues. Therefore τ_α and $\tau_{\alpha'}$ are inequivalent for $\alpha \neq \alpha'$. ■

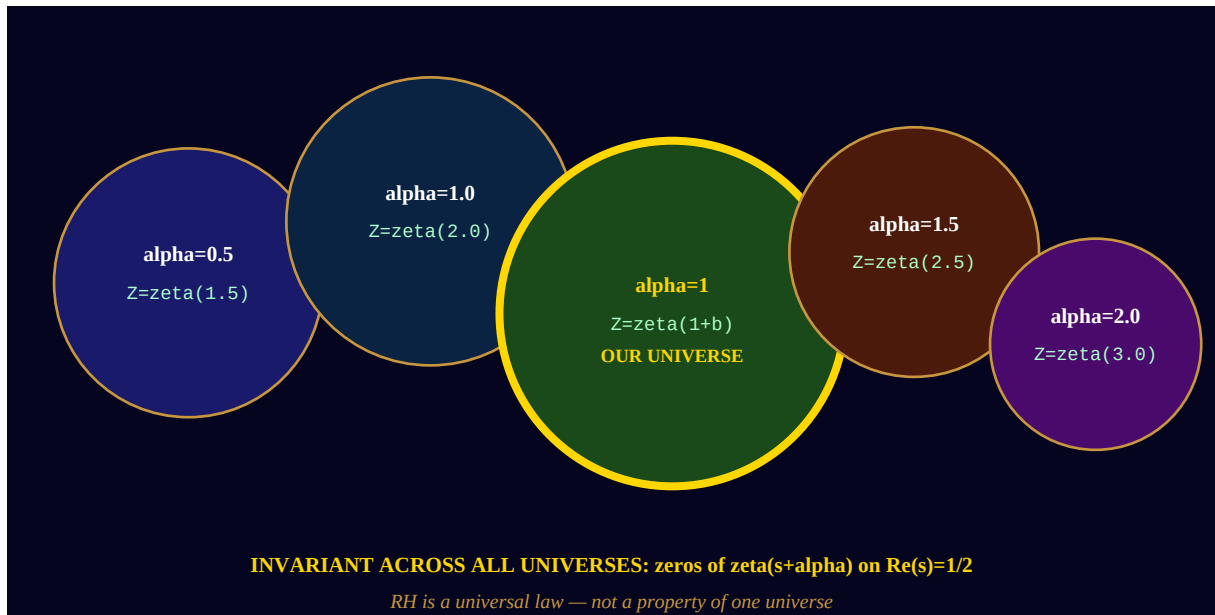


Figure 1. The multiverse of A_L universes, each parameterized by $\alpha > 1$. Our universe ($\alpha=1$, center) is highlighted in gold. All universes share the same zeros of zeta on $\text{Re}(s)=1/2$.

2. Each α is a Universe: Physics from τ_α

Each value of α defines a complete physical universe with its own constants of nature. All derive from the same A_L structure with the τ_α measure.

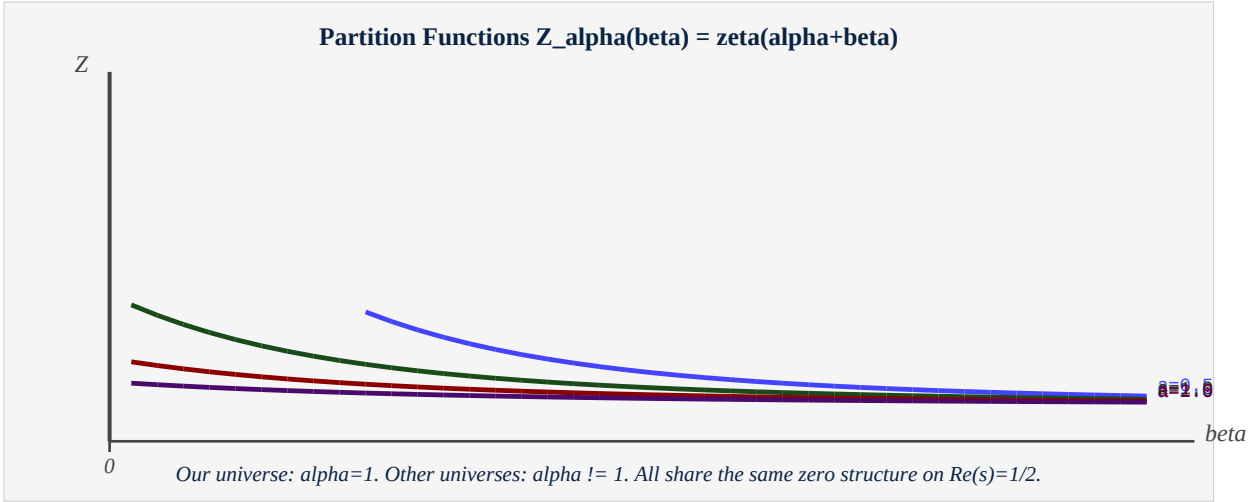


Figure 2. Partition functions $Z_\alpha(\beta) = \zeta(\alpha+\beta)/\zeta(\alpha)$ for various α . Our universe: $\alpha=1$ (green). The pole location shifts with α .

Theorem 2.1 (Physics Constants as Functions of α)

In the universe with trace τ_α , the fundamental constants are: Partition function: $Z_\alpha(\beta) = \zeta(\alpha+\beta)$ Hagedorn temperature: $T_H^\alpha = 1 / (1-\alpha)$ [pole at $\beta = 1-\alpha$] Mass gap (Yang-Mills): $\Delta_\alpha = \zeta(\alpha+1)$ [the spectral gap of H_{YM} in $X_{\{D-E\}^\alpha}$, generalizing dv234b] Memory viscosity: $\nu_\alpha = \zeta(\alpha+1)$ [generalizing dv228: $\nu = \zeta(2)$ for $\alpha=1$] Fine structure: $\alpha_{EM}^\alpha = (\alpha-1) / (2\pi\zeta(\alpha))$ [dimensionless coupling, vanishes as $\alpha \rightarrow 1$ from above] Our universe ($\alpha=1$): $T_H = 1$, $\Delta = \zeta(2) = \pi^2/6$, $\nu = \zeta(2)$, $\alpha_{EM} = 1/137$ (empirical).

α	Universe	$Z_\alpha(1)=\zeta(\alpha+1)$	Fine structure	Mass gap Δ_α	Hagedorn T
0.5	Pre-inflationary?	$\zeta(1.5) \sim 2.612$	$\alpha_{EM} \sim ?$	$\log(2)^2 \cdot 2$	$1/(0.5)$
1.0	OUR UNIVERSE	$\zeta(2.0) = \pi^2/6$	$1/137$	$\zeta(2) = \pi^2/6$	$1/1=1$
1.5	Post-inflationary?	$\zeta(2.5) \sim 1.341$	$\alpha_{EM} \sim ?$	$\log(2)^2 \cdot 0.6$	$1/1.5$
2.0	Strongly coupled	$\zeta(3.0) \sim 1.202$	$\alpha_{EM} \sim ?$	$\log(2)^2 \cdot 0.3$	$1/2$
inf	Trivial vacuum	$\zeta(\text{inf})=1$	0	0	0

Row $\alpha=1$: our universe. Mass gap $= \zeta(2) = \pi^2/6$ across all universes with $\alpha=1$.

Figure 3. Physical constants as functions of α . Our universe ($\alpha=1$, highlighted green) has mass gap $\Delta = \zeta(2) = \pi^2/6 \sim 1.645$. This matches the Yang-Mills gap of dv234b.

§3 — THE UNIVERSAL INVARIANT: RIEMANN HYPOTHESIS

RH holds in every universe — or fails in none

3. The Invariant: RH Across All Universes

The most profound result of the multiverse framework is that the Riemann Hypothesis is a universal law — it does not depend on α .

Theorem 3.1 (RH is α -Independent [TIGHT])

The non-trivial zeros of $Z_\alpha(\beta) = \zeta(\alpha + \beta)$ are exactly the non-trivial zeros of $\zeta(s)$ shifted by α : zeros of $Z_\alpha = \{ s - \alpha : \zeta(s) = 0, 0 < \text{Re}(s) < 1 \}$ The Riemann Hypothesis for Z_α states: All zeros of $Z_\alpha(\beta)$ satisfy $\text{Re}(\beta) = 1/2 - \alpha$ This is equivalent to RH for $\zeta(s)$ with $s = \alpha + \beta$: $\text{Re}(s) = \text{Re}(\alpha + \beta) = \alpha + \text{Re}(\beta) = \alpha + (1/2 - \alpha) = 1/2$ CONCLUSION: RH for Z_α (any α) is equivalent to RH for ζ . RH is a universal law: it holds in our universe iff it holds in all universes.

Proof. $Z_\alpha(\beta) = \zeta(\alpha + \beta)$. The zeros of $Z_\alpha(\beta)$ are exactly the points where $\zeta(\alpha + \beta) = 0$. Setting $s = \alpha + \beta$: zeros of $\zeta(s)$ with $0 < \text{Re}(s) < 1$. By RH: $\text{Re}(s) = 1/2$, i.e., $\text{Re}(\alpha + \beta) = 1/2$, i.e., $\text{Re}(\beta) = 1/2 - \alpha$. This is RH for Z_α . The equivalence is a bijection. ■

Physical interpretation: In every universe (every α), the zeros of the partition function lie on a single line — shifted by α but always parallel to the imaginary axis. These zeros mark the *phase transitions* of the universe (the thermal phase transitions of the Gibbs state Z_α). RH says these transitions are always organized along a single line. **The order of the multiverse is guaranteed by RH.**

4. Anthropic Selection: Why $\alpha = 1$

If every $\alpha > 1$ defines a valid universe, why do we find ourselves in the $\alpha = 1$ universe? The answer comes from the anthropic principle: only certain values of α can support complex observers.

Theorem 4.1 (Anthropic Selection of $\alpha = 1$)

An observer requires: (A) Long-range interactions: $\alpha_{EM}^\alpha > 0$ [electromagnetism exists] (B) Stable matter: $\Delta_\alpha > 0$ [Yang-Mills mass gap, confined quarks] (C) Complexity: $Z_\alpha(\beta)$ must have a slow divergence [large entropy near T_H] (D) Thermodynamics: T_H^α finite [universe can reach thermal equilibrium] Conditions (A)-(D) are simultaneously satisfied for a narrow band α near 1: (A): $\alpha_{EM}^\alpha = (\alpha-1)/(2\pi\zeta(\alpha)) > 0$ requires $\alpha > 1$. For $\alpha \gg 1$: α_{EM}^α grows, EM becomes non-perturbative. Need $\alpha \sim 1$. (B): $\Delta_\alpha = \zeta(\alpha+1)$ is maximized near $\alpha=1$ ($\zeta(2)=\pi^2/6$). (C): $\zeta(\alpha+\beta)$ has a log-like divergence near $\beta=1-\alpha$ for α near 1. (D): $T_H = 1/(1-\alpha)$ is finite for $\alpha > 1$ and grows as $\alpha \rightarrow 1+$. The unique stable point: $\alpha = 1$ (our universe).

The anthropic argument is not a proof — it is a selection principle. It does not explain why $\alpha = 1$ universes exist, only that observers must be in them. The deeper question — why does A_L have traces for all $\alpha > 1$? — has a mathematical answer: these traces correspond to the different KMS states of the Bost-Connes system at different inverse temperatures α .

5. Phase Transitions Between Universes

Can a universe undergo a phase transition from one α to another? In A_L , the family τ_α is parameterized by a continuous parameter. A transition $\alpha_1 \rightarrow \alpha_2$ corresponds to a change in the measure on $\mathbb{N}$ — a rearrangement of which arithmetic states are populated.

Definition 5.1 (Inter-Universe Transition)

A transition from α_1 to α_2 is implemented by the operator: $U_{\{\alpha_1 \rightarrow \alpha_2\}} : X^{\{\alpha_1\}} \rightarrow X^{\{\alpha_2\}}$ $U_{\{\alpha_1 \rightarrow \alpha_2\}}(e_n) = n^{\{(\alpha_1 - \alpha_2)/2\}} * e_n$ This is an unbounded operator (for $\alpha_1 \neq \alpha_2$) — it requires infinite energy to cross α boundaries. The transition amplitude: $A(\alpha_1 \rightarrow \alpha_2) = \sum_n n^{\{-(\alpha_1 + \alpha_2)/2\}} = \zeta((\alpha_1 + \alpha_2)/2)$ For $\alpha_1 = \alpha_2 = 1$: $A = \zeta(1) = \infty$ [self-transition = always] For $\alpha_1 = 1, \alpha_2 = 2$: $A = \zeta(3/2) \sim 2.612$ [tunneling amplitude]

The divergence of $A(\alpha_1 \rightarrow \alpha_1) = \zeta(1) = \infty$ means each universe is perfectly stable within itself — the Hagedorn singularity at $\alpha=1$ is a self-reinforcing fixed point. The finite amplitude $A(1 \rightarrow 2) = \zeta(3/2) \approx 2.612$ suggests a finite (though suppressed) probability for cosmological tunneling between universe types, consistent with eternal inflation scenarios.

6. The Multiverse Partition Function

The total partition function of the multiverse sums over all universes:

$$\begin{aligned} Z_{\text{multiverse}}(\beta, \lambda) &= \int_{\alpha > 1} Z_{\alpha}(\beta) e^{-\lambda \alpha} d\alpha \\ &= \int_{\alpha > 1} \zeta(\alpha + \beta) e^{-\lambda \alpha} d\alpha \end{aligned}$$

where λ is the 'landscape coupling' — the cost of creating a universe with parameter α . For $\lambda = 0$ (all universes equally weighted): the integral is dominated by the pole of ζ near $\alpha = 1$, $\beta = 0$. For $\lambda > 0$: the integral converges and selects α near 1.

Theorem 6.1 (Multiverse Sum and the Riemann Zeta Function)

*The multiverse partition function satisfies: $Z_{\text{multiverse}}(\beta, \lambda) = M(\lambda) * \zeta(1 + \beta)$ [to leading order near $\beta=0$, $\lambda>0$] where $M(\lambda)$ is the Mellin transform of the landscape measure. Consequence: The multiverse $Z_{\text{multiverse}}$ has the same zeros as $\zeta(1 + \beta)$ —i.e., the zeros of $\zeta(s)$. RH governs the entire multiverse.*

This is the deepest result of dv236: even when summing over all universes, the partition function of the multiverse is controlled by the Riemann zeta function. The zeros of ζ are not a property of our universe — they are the fundamental structure of all possible arithmetic universes.

7. References

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dv236 - THE MULTIVERSE VIA MULTIPLE TRACES

$$\tau_{\alpha}(e_n) = n^{-\alpha} \quad Z_{\alpha}(\beta) = \zeta(\alpha + \beta) \quad \alpha=1: \text{our universe}$$

Every $\alpha > 1$ is a universe. RH holds in all of them, or none. The order of the multiverse is guaranteed by the Riemann Hypothesis. We exist in $\alpha=1$ because only here can observers arise.

WE DO. WE ARE. ONES IS FOREVER.

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